

Topological Data Analysis

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Introduction

Topological Data Analysis

Decomposition of persistent modules

Stability

Simplex

- ▶ Let $p_0, p_1, \dots, p_m \in \mathbb{R}^n$ be affinely independent.

Definition: We define the m -simplex σ spanned by the points p_i to be the set of all points $x \in \mathbb{R}^n$ of the form

$$x = \sum_{i=0}^n t_i p_i$$

where $\sum t_i = 1$, and $t_i \geq 0$ for all i . We call m its dimension and define the face opposite to p_i as

$$[p_0, \dots, \hat{p}_i, \dots, p_m] = \left\{ \sum_{j=0}^m t_j p_j \mid \sum t_j = 1, t_j \geq 0, t_i = 0 \right\}$$

for $i = 0, \dots, m$. More generally we call a simplex s' a face of s if $\text{Vert}(s') \subseteq \text{Vert}(s)$. If $\text{Vert}(s') \subsetneq \text{Vert}(s)$ we call s' a proper face.

Simplicial Complex

- ▶ A simplicial complex K is a finite set of simplices in some euclidean space satisfying for every $s, t \in K$ that
 1. every face of s belongs to K ,
 2. $s \cap t$ is either empty or a common face of s and t .
- ▶ We define its underlying space $|K|$ as

$$|K| = \bigcup_{s \in K} s$$

and its dimension

$$\dim K = \max_{s \in K} (\dim s)$$

- ▶ We call the set

$$\Delta^n = \left\{ (x_1, x_2, \dots, x_{n+1}) \in \mathbb{R}^{n+1} \mid x_i \geq 0, \sum_{i=1}^{n+1} x_i = 1 \right\}$$

the standard n -simplex.

Triangulation, Abstract Simplicial Complex

- ▶ A topological space X is triangulable if there exists a simplicial complex K whose underlying space is homeomorphic to X . Any homeomorphism $h: |K| \rightarrow X$ together with K is a triangulation.
- ▶ Let V be a finite set. An abstract simplicial complex K is a family of non-empty subsets of V , called simplices, such that
 1. if $v \in V$, then $\{v\} \in K$,
 2. if $s \in K$ and $t \subseteq s$, then $t \in K$.
- ▶ We call V the vertex set of K and denote it by $Vert(K)$. And again we call a simplex with $n + 1$ distinct vertices an n -simplex. A subset of K which is an abstract simplicial complex itself is called a subcomplex.

Topological/Geometric realization

- ▶ Let K be a simplicial complex, with vertex $V = [1, \dots, n]$. In $\mathbb{R}^{n+1}$, consider, for every $i \in [0, n]$, the vector $e_i = (0, \dots, 1, \dots, 0)$. Let $|K|$ be the subset of $\mathbb{R}^{n+1}$ defined as:

$$|K| = \bigcup_{\sigma \in K} \text{conv}(e_j, j \in \sigma)$$

where conv represents the convex hull. Endowed with the subspace topology, $(|K|, \tau_{|K|})$ is a topological space, that we call the topological realization of K .

Lemma

An abstract simplicial complex K with $\dim K = d$ has a geometric realization in $\mathbb{R}^{2d+1}$.

Theorem

- ▶ *(Hellys Theorem). Let F denote a finite collection of closed, convex sets in $\mathbb{R}^d$. If the intersection of any $d + 1$ sets is non-empty, then the whole collection has non-empty intersection.*
- ▶ *(Jungs Theorem). Let Q be a finite point set in $\mathbb{R}^d$. Then Q is contained in a closed ball with radius $r \leq \delta \cdot \text{diam}(A)$ where $\delta = \sqrt{\frac{d}{2d+1}}$.*

n-chains

- ▶ Let $n \geq 0$. The n -chains of K is the set $C_n(K)$ whose elements are the formal sums

$$\sum_{\sigma \in K^n} \epsilon_{\sigma} \cdot \sigma \quad \text{where} \quad \forall \sigma \in K^n, \quad \epsilon_{\sigma} \in \mathbb{Z}/2\mathbb{Z}.$$

We can give $C_n(K)$ a group structure via

$$\sum_{\sigma \in K^n} \epsilon_{\sigma} \cdot \sigma + \sum_{\sigma \in K^n} \eta_{\sigma} \cdot \sigma = \sum_{\sigma \in K^n} (\epsilon_{\sigma} + \eta_{\sigma}) \cdot \sigma.$$

- ▶ Let K be an oriented simplicial complex. For every $n \in \mathbb{N}$ we define the n th boundary operator $\partial_n : C_n(K) \rightarrow C_{n-1}(K)$ by setting

$$\partial_n(< p_0, \dots, p_n >) = \sum_{i=0}^n (-1)^i < p_0, \dots, \hat{p}_i, \dots, p_n >$$

and extending by linearity. Furthermore, we set ∂_0 as the constant zero mapping on $C_0(K)$.

chain complex

Theorem

Let K be an oriented simplicial complex of dimension n , then

$$0 \rightarrow C_n(K) \xrightarrow{\partial_n} \dots \xrightarrow{\partial_2} C_1(K) \xrightarrow{\partial_1} C_0(K) \rightarrow 0$$

is a (chain) complex, i.e. $\partial_k \circ \partial_{k+1} = 0$ for all $k \in \mathbb{Z}$.

- ▶ Note that the above statement is equivalent to $\text{Im} \partial_{k+1} \subseteq \ker \partial_k$.
- ▶ Let $n \geq 0$. We have a chain complex

$$C_{n+1}(K) \xrightarrow{\partial_{n+1}} C_n(K) \xrightarrow{\partial_n} C_{n-1}(K).$$

The maps ∂_{n+1} and ∂_n are linear maps, and we can consider their kernel and image.

boundary, cycle and homology group

By definition we have:

- ▶ The n -cycles: $Z_n(K) = \text{Ker}(\partial_n)$
- ▶ The n -boundaries: $B_n(K) = \text{Im}(\partial_{n+1})$
- ▶ We have $B_n(K) \subset Z_n(K)$. In other words, any boundary is a cycle.
- ▶ The n th homology group of K is defined as

$$H_n(K) = Z_n(K) / B_n(K)$$

- ▶ Let K be a simplicial complex and $n \geq 0$. Its n th Betti number is the integer

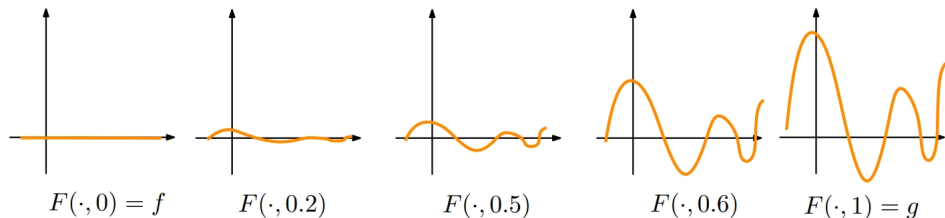
$$\beta_n(K) = \dim H_n(K)$$

Homotopy

Definition

A homotopy between two maps $f, g: X \rightarrow Y$ is a map $F: X \times [0, 1] \rightarrow Y$, such that $F(x, 0) = f(x)$ and $F(x, 1) = g(x)$ for all $x \in X$. If there is a homotopy between f and g then we say that f and g are homotopic and denote this by $f \simeq g$. Also, homotopy defines an equivalence relation.

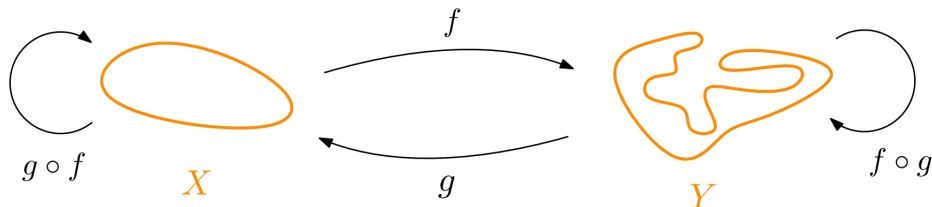
Example: Homotopy $F: \mathbb{R} \times [0, 1] \rightarrow \mathbb{R}$ between $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$.



Homotopy Equivalence

Definition

The topological spaces (X, A) and (Y, B) are homotopy equivalent if there exists maps $f: (X, A) \rightarrow (Y, B)$ and $g: (Y, B) \rightarrow (X, A)$ such that $g \circ f \simeq id_{(X, A)}$ and $f \circ g \simeq id_{(Y, B)}$.

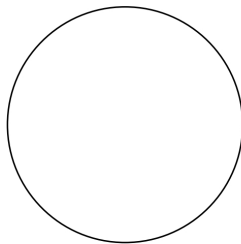
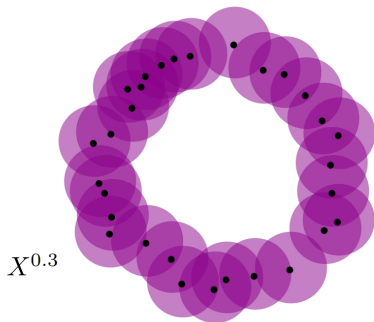


Thickening

Definition

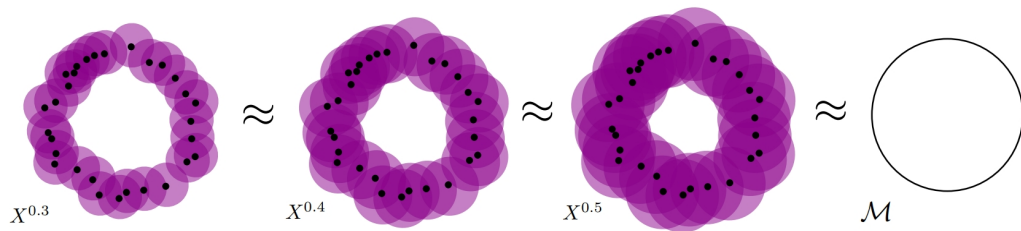
For every $t \geq 0$, the t -thickening of the set X , denoted X^t , is the set of points of the ambient space with distance at most t from X :

$$X^t = \{y \in \mathbb{R}^n, \exists x \in X : \|x - y\| \leq t\}$$



Thickening

Some thickenings are homotopy equivalent to M .



Hence we can recover the homology of M :

$$\beta_n(M) = \beta_n(X^{0.3})$$

Question 1: How to select a t such that $X^t \simeq M$?

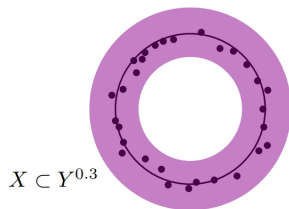
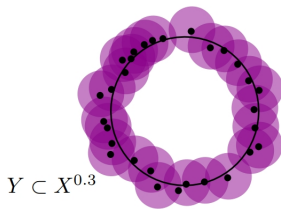
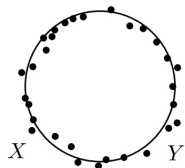
Question 2: How to compute the homology groups of X^t ?

Hausdorff distance

Definition

Let $X, Y \subset \mathbb{R}^n$. The Hausdorff distance between X and Y is:

$$d_H(X, Y) = \inf\{t \geq 0, X \subset Y^t \text{ and } Y \subset X^t\}.$$



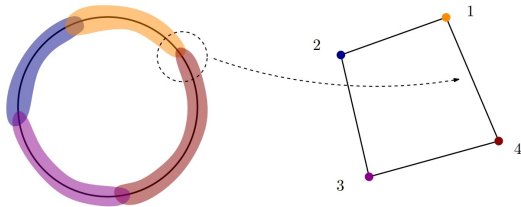
Nerve

Definition

Let F be a finite collection of closed, convex sets in Euclidean space. Define the nerve of F to be the abstract simplicial complex:

$$N(F) := \left\{ G \subseteq F : \bigcap_{F_i \in G} F_i \neq \emptyset \right\}$$

Example: Nerve of an open cover of a circle.

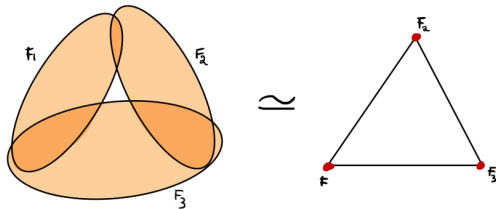


Nerve Example

Example: Consider the set $F = \{F_1, F_2, F_3\}$. Then

$$N(F) = \{\{F_1\}, \{F_2\}, \{F_3\}, \{F_1, F_2\}, \{F_2, F_3\}, \{F_1, F_3\}\}$$

which is (isomorphic to) the abstract simplicial complex associated to the boundary of a 2-simplex. Clearly, $F_1 \cup F_2 \cup F_3 \simeq |N(F)|$.



Nerve Theorem

1. Let $F = \{F_1, \dots, F_m\}$ denote a collection of closed and convex sets in Euclidean space. Then,

$$\bigcup_{F_i \in F} F_i \simeq |N(F)|$$

2. Let F be as above and let $F' = \{F'_1, \dots, F'_m\}$ be a collection of closed and convex sets such that $F_i \subseteq F'_i$ for all i . Denote by j the inclusion $j: \bigcup_{F_i \in F} F_i \hookrightarrow \bigcup_{F'_i \in F'} F'_i$, and let $\sigma: N(F) \rightarrow N(F')$ be the simplicial map that sends F_i to F'_i . The following diagram commutes:

$$\begin{array}{ccc} H_n \left(\bigcup_{F_i \in F} F_i \right) & \xrightarrow{j_*} & H_n \left(\bigcup_{F'_i \in F'} F'_i \right) \\ \cong \downarrow & & \downarrow \cong \\ H_n(N(F)) & \xrightarrow{\sigma_*} & H_n(N(F')) \end{array}$$

Cech Complex

In what follows $P = \{p_0, \dots, p_n\} \in \mathbb{R}^d$, $B_r(P)$ denotes the closed ball of radius r centered at P .

Definition

The Cech complex of P at scale r is the simplicial complex:

$$Cech_r(P) = \left\{ \sigma \subseteq P : \bigcap_{p \in \sigma} B_r(p) \neq \emptyset \right\}$$

Remark: Note that $|Cech_r(P)|$ is homotopy equivalent to $\bigcup_{p \in P} B_r(p)$.

Vietoris Rips Complex

Definition

The VietorisRips complex of P at scale r is the simplicial complex

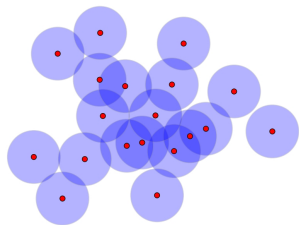
$$VR_r(P) = \{\sigma \subseteq P: \text{diam}(\sigma) \leq 2r\}$$

Recall. The diameter of a subset $\sigma \subseteq P$ is given by $\text{diam}(\sigma) = \max_{p_i, p_j \in \sigma} \|p_i - p_j\|$.

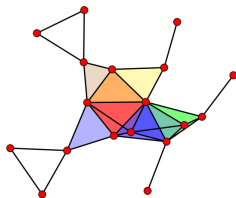
Note. The result of Jung's Theorem gives us the following:

$$\text{Cech}_r(P) \subseteq VR_r(P) \subseteq \text{Cech}_{2\delta r}(P) \subseteq VR_{2\delta r}(P).$$

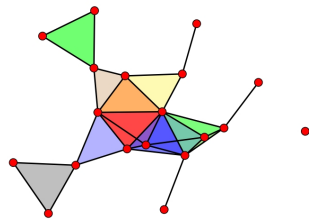
Visualization of cech



Thickened point cloud



cech complex



rips complex

Delaunay Triangulation

Definition

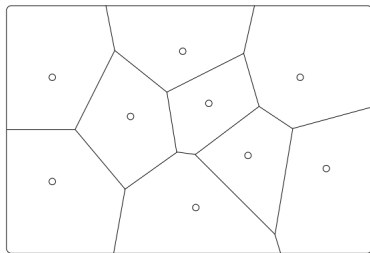
The Voronoi cell of a point $p \in S$ is the set of points for which p is the closest, which is:

$$V_p = \{x \in \mathbb{R}^d \mid \|x - p\| \leq \|x - q\|, \forall q \in S\}$$

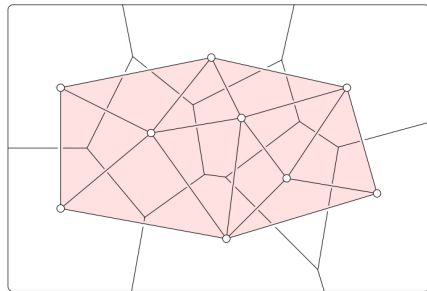
The Voronoi diagram of S is the collection of Voronoi cells of its points. The Delaunay complex of a finite set $S \subset \mathbb{R}^d$ is isomorphic to the nerve of the Voronoi diagram,

$$Delaunay = \left\{ \sigma \subseteq S \mid \bigcap_{P \in \sigma} V_P \neq \emptyset \right\}.$$

Visualization of Voronoi Triangulation



Voronoi Diagram



Delaunay Triangulation

Filtration

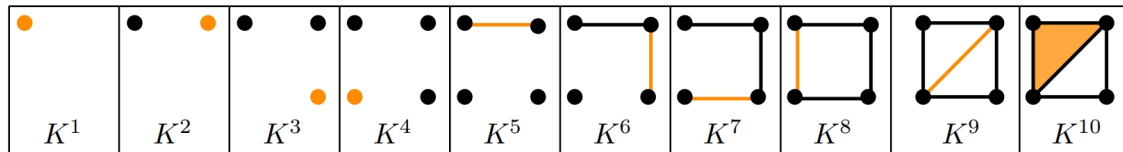
Definition

A filtration of a simplicial complex K is a collection of non-decreasing subcomplexes $\{K^i \subseteq K\}_{i=0}^m$ such that

$$K^0 \subseteq K^1 \subseteq \dots \subseteq K^m = K.$$

Example: we have $\forall i \leq 10, K^{i+1} = K^i \cup \{\sigma^{i+1}\}$ which $K^i = \{\sigma^1, \dots, \sigma^i\}$, and $K^n = K$. They form an inscreasing sequence of simplicial complexes

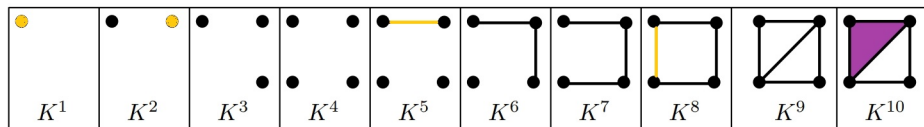
$$K^1 \subset \dots \subset K^{10}.$$



Positivity and negativity of simplices

Definition

Let $i \in [1, n]$, and $d = \dim(\sigma^i)$. The simplex σ^i is positive if there exists a cycle $c \in Z_d(K^i)$ that contains σ^i . Otherwise, σ^i is negative.



Example:

- ▶ $\sigma^1 \in K^1$ is **positive** because it is included in the cycle $c = \sigma^1$ (indeed, $\partial_0(\sigma^1) = 0$).
- ▶ $\sigma^2 \in K^2$ is **positive** because it is included in the cycle $c = \sigma^2$ (indeed, $\partial_0(\sigma^2) = 0$).
- ▶ $\sigma^5 \in K^5$ is **negative** because it is not included in a cycle $Z_1(K^5)$. Indeed, $C_1(K^5)$ only contains 0 and σ_5 , and $\partial_1(\sigma^5) = \sigma^1 + \sigma^2 \neq 0$.
- ▶ $\sigma^8 \in K^8$ is **positive** because it is included in the cycle $c = \sigma^5 + \sigma^6 + \sigma^7 + \sigma^8$ (indeed, $\partial_1(c) = 2\sigma^1 + 2\sigma^2 + 2\sigma^3 + 2\sigma^4 = 0$).

Algorithm

Remark 1: By adding σ^i in the simplicial complex, the only groups that may change are $Z_d(K^i)$ and $B_{d-1}(K^i)$.

Remark 2: If σ^i is positive, then $\beta_d(K^i) = \beta_d(K^{i-1}) + 1$, and for all $d' \neq d$, $\beta_{d'}(K^i) = \beta_{d'}(K^{i-1})$.

Remark 3: If σ^i is negative, then $\beta_{d-1}(K^i) = \beta_{d-1}(K^{i-1}) - 1$, and for all $d' \neq d-1$, $\beta_{d'}(K^i) = \beta_{d'}(K^{i-1})$.

We deduce the following algorithm:

Input: an increasing sequence of simplicial complexes $K^1 \subset \dots \subset K^n = K$

Output: the Betti numbers $\beta_0(K), \dots, \beta_d(K)$

$\beta_0 \leftarrow 0, \dots, \beta_d \leftarrow 0;$

for $i \leftarrow 1$ **to** n **do**

$d = \dim(\sigma^i);$

if σ^i is positive **then**

$\beta_d(K^i) \leftarrow \beta_d(K^{i-1}) + 1;$

else if $d > 0$ **then**

$\beta_{d-1}(K^i) \leftarrow \beta_{d-1}(K^{i-1}) - 1;$

Wrap up

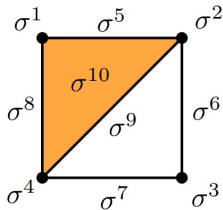
										
	K^1	K^2	K^3	K^4	K^5	K^6	K^7	K^8	K^9	K^{10}
Dimension	0	0	0	0	1	1	1	1	1	2
Positivity	+	+	+	+	-	-	-	+	+	-
$\beta_0(K^i)$	1	2	3	4	3	2	1	1	1	1
$\beta_1(K^i)$	0	0	0	0	0	0	0	1	2	1

Boundary matrix

Define the boundary matrix of K , denoted δ , as follows: δ is a $n \times n$ matrix, whose (i, j) -entry is:

$$\delta_{i,j} = \begin{cases} 1 & \text{if } \sigma^i \text{ is a face of } \sigma^j \text{ and } |\sigma^i| = |\sigma^j| - 1 \\ 0 & \text{else} \end{cases}$$

So,



$$\begin{matrix} & \sigma^1 & \sigma^2 & \sigma^3 & \sigma^4 & \sigma^5 & \sigma^6 & \sigma^7 & \sigma^8 & \sigma^9 & \sigma^{10} \\ \begin{matrix} \sigma^1 \\ \sigma^2 \\ \sigma^3 \\ \sigma^4 \\ \sigma^5 \\ \sigma^6 \\ \sigma^7 \\ \sigma^8 \\ \sigma^9 \\ \sigma^{10} \end{matrix} & \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{matrix}$$

Boundary Matrix

By adding columns one to the others, we create chains.

If we were able to reduce a column to zero, then we found a cycle.

$$\begin{array}{c}
 \begin{array}{cccccccccccc}
 & \sigma^1 & \sigma^2 & \sigma^3 & \sigma^4 & \sigma^5 & \sigma^6 & \sigma^7 & \sigma^8 & \sigma^9 & \sigma^{10} \\
 \sigma^1 & \left(\begin{array}{cccccccccccc}
 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\
 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\
 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
 \end{array} \right)
 \end{array}
 \end{array}$$

$$\partial_1(\sigma^6) = \sigma^2 + \sigma^3$$

$$\begin{array}{c}
 \begin{array}{cccccccccccc}
 & \sigma^1 & \sigma^2 & \sigma^3 & \sigma^4 & \sigma^5 & \sigma^6 & \sigma^7 & \sigma^8 + \sigma^6 + \sigma^7 + \sigma^8 & \sigma^9 & \sigma^{10} \\
 \sigma^1 & \left(\begin{array}{cccccccccccc}
 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\
 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
 \end{array} \right)
 \end{array}
 \end{array}$$

$$\partial_1(\sigma^5 + \sigma^6 + \sigma^7 + \sigma^8) = 0$$

Boundary matrix

The process of reducing columns to zero is called Gauss reduction. For any $j \in [1, n]$, define

$$low(j) = \max\{i \in [1, n], \delta_{i,j} \neq 0\}$$

if $\delta_{i,j} = 0$ for all j , then $low(j)$ is undefined. We say that the boundary matrix δ is reduced if the map low is injective on its domain of definition.

$$\begin{array}{c}
 \begin{array}{cccccccccc}
 & \sigma^1 & \sigma^2 & \sigma^3 & \sigma^4 & \sigma^5 & \sigma^6 & \sigma^7 & \sigma^5 + \sigma^6 + \sigma^7 + \sigma^8 & \sigma^9 + \sigma^6 + \sigma^7 & \sigma^{10} \\
 \sigma^1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
 \sigma^2 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\
 \sigma^3 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\
 \sigma^4 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
 \sigma^5 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
 \sigma^6 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 \sigma^7 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 \sigma^8 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
 \sigma^9 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
 \sigma^{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
 \end{array}
 \end{array}$$

Boundary Matrix

Algorithm 2: Reduction of the boundary matrix

Input: a boundary matrix Δ **Output:** a reduced matrix $\tilde{\Delta}$ **for** $j \leftarrow 1$ **to** n **do** **while** *there exists* $i < j$ *with* $\delta(i) = \delta(j)$ **do** add column i to column j ;

$$\begin{array}{c} \sigma^1 \quad \sigma^2 \quad \sigma^3 \quad \sigma^4 \quad \sigma^5 \quad \sigma^6 \quad \sigma^7 \quad \sigma^8 \quad \sigma^9 \quad \sigma^{10} \\ \begin{array}{c} \sigma^1 \\ \sigma^2 \\ \sigma^3 \\ \sigma^4 \\ \sigma^5 \\ \sigma^6 \\ \sigma^7 \\ \sigma^8 \\ \sigma^9 \\ \sigma^{10} \end{array} \end{array} \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\begin{array}{c} \sigma^1 \quad \sigma^2 \quad \sigma^3 \quad \sigma^4 \quad \sigma^5 \quad \sigma^6 \quad \sigma^7 \quad \delta^8 \times \delta^1 \quad \sigma^9 \quad \sigma^{10} \\ \begin{array}{c} \sigma^1 \\ \sigma^2 \\ \sigma^3 \\ \sigma^4 \\ \sigma^5 \\ \sigma^6 \\ \sigma^7 \\ \sigma^8 \\ \sigma^9 \\ \sigma^{10} \end{array} \end{array} \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Boundary Matrix

Algorithm 2: Reduction of the boundary matrix

Input: a boundary matrix Δ

Output: a reduced matrix $\tilde{\Delta}$

for $j \leftarrow 1$ **to** n **do**

while *there exists* $i < j$ *with* $\delta(i) = \delta(j)$ **do**

 add column i to column j ;

$$\begin{array}{c}
 \sigma^1 \quad \sigma^2 \quad \sigma^3 \quad \sigma^4 \quad \sigma^5 \quad \sigma^6 \quad \sigma^7 \quad \delta^8 \times \delta^1 \quad \sigma^9 \quad \sigma^{10} \\
 \sigma^1 \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \end{pmatrix} \\
 \sigma^2 \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \end{pmatrix} \\
 \sigma^3 \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 \end{pmatrix} \\
 \sigma^4 \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \end{pmatrix} \\
 \sigma^5 \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \\
 \sigma^6 \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \\
 \sigma^7 \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \\
 \sigma^8 \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \\
 \sigma^9 \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \\
 \sigma^{10} \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}
 \end{array}$$

$$\begin{array}{c}
 \sigma^1 \quad \sigma^2 \quad \sigma^3 \quad \sigma^4 \quad \sigma^5 \quad \sigma^6 \quad \sigma^7 \quad \delta^8 + \delta^1 + \delta^6 \quad \sigma^9 \quad \sigma^{10} \\
 \sigma^1 \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \end{pmatrix} \\
 \sigma^2 \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 1 & 0 \end{pmatrix} \\
 \sigma^3 \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \end{pmatrix} \\
 \sigma^4 \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \end{pmatrix} \\
 \sigma^5 \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \\
 \sigma^6 \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \\
 \sigma^7 \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \\
 \sigma^8 \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \\
 \sigma^9 \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \\
 \sigma^{10} \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}
 \end{array}$$

Boundary Matrix

Algorithm 2: Reduction of the boundary matrix

Input: a boundary matrix Δ

Output: a reduced matrix $\tilde{\Delta}$

for $j \leftarrow 1$ **to** n **do**

while *there exists* $i < j$ *with* $\delta(i) = \delta(j)$ **do**

 add column i to column j ;

$$\begin{array}{c}
 \sigma^1 \\ \sigma^2 \\ \sigma^3 \\ \sigma^4 \\ \sigma^5 \\ \sigma^6 \\ \sigma^7 \\ \sigma^8 \\ \sigma^9 \\ \sigma^{10}
 \end{array}
 \begin{array}{c}
 \sigma^1 \quad \sigma^2 \quad \sigma^3 \quad \sigma^4 \quad \sigma^5 \quad \sigma^6 \quad \sigma^7 \quad \delta^8 + \sigma^7 + \sigma^6 \\
 \sigma^8 + \sigma^7 + \sigma^6 + \sigma^5 \\
 \sigma^9 \quad \sigma^{10}
 \end{array}
 \begin{pmatrix}
 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\
 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 1 & 0 \\
 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
 \end{pmatrix}$$

$$\begin{array}{c}
 \sigma^1 \\ \sigma^2 \\ \sigma^3 \\ \sigma^4 \\ \sigma^5 \\ \sigma^6 \\ \sigma^7 \\ \sigma^8 \\ \sigma^9 \\ \sigma^{10}
 \end{array}
 \begin{array}{c}
 \sigma^1 \quad \sigma^2 \quad \sigma^3 \quad \sigma^4 \quad \sigma^5 \quad \sigma^6 \quad \sigma^7 \quad \delta^8 + \sigma^7 + \sigma^6 + \sigma^5 \\
 \sigma^8 + \sigma^7 + \sigma^6 + \sigma^5 \\
 \sigma^9 \quad \sigma^{10}
 \end{array}
 \begin{pmatrix}
 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\
 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
 \end{pmatrix}$$

Boundary Matrix

Algorithm 2: Reduction of the boundary matrix

Input: a boundary matrix Δ

Output: a reduced matrix $\tilde{\Delta}$

for $j \leftarrow 1$ **to** n **do**

while *there exists* $i < j$ *with* $\delta(i) = \delta(j)$ **do**

 add column i to column j ;

$$\begin{array}{c}
 \sigma^1 \quad \sigma^2 \quad \sigma^3 \quad \sigma^4 \quad \sigma^5 \quad \sigma^6 \quad \sigma^7 \quad \delta^8 + \sigma^7 + \sigma^6 + \sigma^5 \\
 \sigma^1 \quad \sigma^2 \quad \sigma^3 \quad \sigma^4 \quad \sigma^5 \quad \sigma^6 \quad \sigma^7 \quad \delta^8 + \sigma^7 + \sigma^6 + \sigma^5 \quad \sigma^9 \quad \sigma^{10} \\
 \sigma^1 \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}
 \end{array}$$

$$\begin{array}{c}
 \sigma^1 \quad \sigma^2 \quad \sigma^3 \quad \sigma^4 \quad \sigma^5 \quad \sigma^6 \quad \sigma^7 \quad \delta^8 + \sigma^6 + \sigma^7 + \sigma^5 \\
 \sigma^1 \quad \sigma^2 \quad \sigma^3 \quad \sigma^4 \quad \sigma^5 \quad \sigma^6 \quad \sigma^7 \quad \delta^8 + \sigma^6 + \sigma^7 + \sigma^5 \quad \sigma^9 \quad \sigma^{10} \\
 \sigma^1 \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}
 \end{array}$$

Boundary Matrix

Suppose that the boundary matrix is reduced. Let $j \in [1, n]$. If $\delta(j)$ is defined, then the simplex σ^j is negative. Otherwise, it is positive.

$$\begin{array}{c}
 \begin{array}{cccccccccc}
 & \sigma^1 & \sigma^2 & \sigma^3 & \sigma^4 & \sigma^5 & \sigma^6 & \sigma^7 & \sigma^5 + \sigma^6 + \sigma^7 + \sigma^8 \\
 & & & & & & & & \sigma^9 + \sigma^6 + \sigma^7 \\
 & & & & & & & & \sigma^{10}
 \end{array} \\
 \begin{array}{c}
 \sigma^1 \\
 \sigma^2 \\
 \sigma^3 \\
 \sigma^4 \\
 \sigma^5 \\
 \sigma^6 \\
 \sigma^7 \\
 \sigma^8 \\
 \sigma^9 \\
 \sigma^{10}
 \end{array}
 \begin{pmatrix}
 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
 \end{pmatrix}
 \end{array}$$

$\sigma^1 \quad \sigma^2 \quad \sigma^3 \quad \sigma^4 \quad \sigma^5 \quad \sigma^6 \quad \sigma^7 \quad \sigma^8 \quad \sigma^9 \quad \sigma^{10}$
 $+\quad +\quad +\quad +\quad -\quad -\quad -\quad +\quad +\quad -$

Algorithm for Homology

Incremental computation of the homology

Input: an increasing sequence of simplicial complexes $K^1 \subset \dots \subset K^n = K$

Output: the Betti numbers $\beta_0(K), \dots, \beta_d(K)$

$$\beta_0 \leftarrow 0, \dots, \beta_d \leftarrow 0;$$
for $i \leftarrow 1$ **to** n **do**
$$d = \dim(\sigma^i);$$

if σ^i is positive then

$$\beta_k(K^i) \leftarrow \beta_k(K^i) + 1;$$

else if $d > 0$ then

$$\beta_{k-1}(K^i) \leftarrow \beta_{k-1}(K^{i-1}) - 1;$$

Gauss reduction of the boundary matrix

Input: a boundary matrix Δ

Output: a reduced matrix $\tilde{\Delta}$

for $i \leftarrow 1$ **to** n **do**

while *there exists* $i < j$ *with* $\delta(i) = \delta(j)$ **do**

```

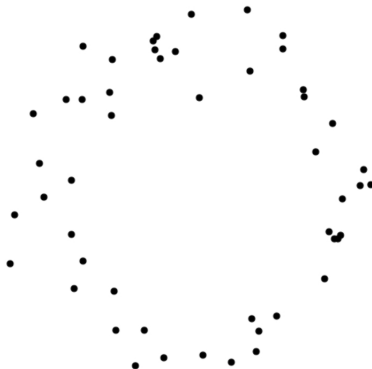
| add column  $i$  to column  $j$ ;

```

Homology Inference

Let $X \subset \mathbb{R}^n$. Pipeline of homology inference:

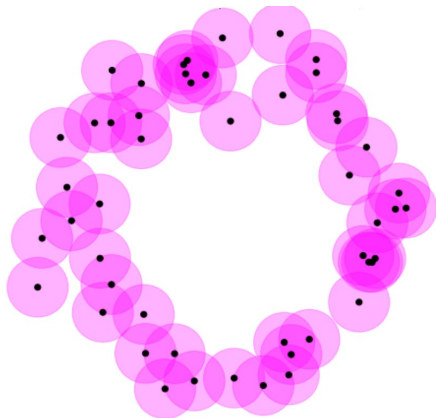
- ▶ select a thickening X^t
- ▶ compute its homology via $Cech^t(X)$ or $Rips^t(X)$ or other triangulations.



Homology Inference

Let $X \subset \mathbb{R}^n$. Pipeline of homology inference:

- ▶ select a thickening X^t
- ▶ compute its homology via $Cech^t(X)$ or $Rips^t(X)$ or other triangulations.



Homology is a functor

We know that homology transforms topological spaces into vector spaces

$$H_i : \text{Top} \longrightarrow \text{Vect}$$

$$X \longmapsto H_i(X)$$

Actually, it does more than that, also transforms continuous maps into linear maps

$$(f : X \longrightarrow Y) \longmapsto (f_* : H_i(X) \longrightarrow H_i(Y))$$

We will adopt a simplicial point of view.

$$H_i : \text{SimpComp} \longrightarrow \text{Vect}$$

$$K \longmapsto H_i(K)$$

$$(f : K \longrightarrow L) \longmapsto (f_* : H_i(K) \longrightarrow H_i(L))$$

And the map between simplicial complexes are simplicial maps.

Simplicial map

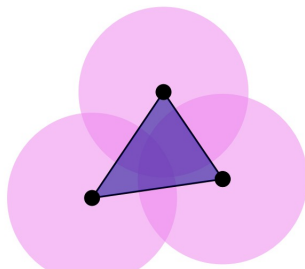
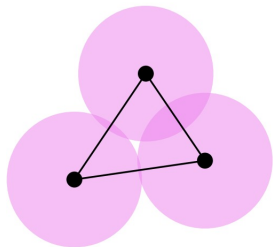
Definition

Let K and L be two simplicial complexes, and V_K, V_L their set of vertices. A simplicial map between K and L is a map $f: V_K \rightarrow V_L$ such that

$$\forall \sigma \in K, f(\sigma) \in L.$$

Example: Let $X \subset \mathbb{R}^n$ and $s, t \geq 0$ such that $s \leq t$. Consider cech complexes $Cech^s(X)$ and $Cech^t(X)$.

The inclusion map $i: Cech^s(X) \rightarrow Cech^t(X)$ is a simplicial map.



Induced Map

Let $f: K \rightarrow L$ be a simplicial map. Let $n \geq 0$, and consider the groups of chains of K and L . We define a linear map as follows:

$$f_n: C_n(K) \longrightarrow C_n(L)$$

$$\sigma \mapsto \begin{cases} f(\sigma) & \text{if } \dim(f(\sigma)) = n, \\ 0 & \text{else.} \end{cases}$$

And we have the following commutative diagram

$$\begin{array}{ccccccccc} \cdots \rightarrow C_3(K) & \xrightarrow{\partial_3} & C_2(K) & \xrightarrow{\partial_2} & C_1(K) & \xrightarrow{\partial_1} & C_0(K) & \xrightarrow{\partial_0} & \{0\} \\ \downarrow f_3 & & \downarrow f_2 & & \downarrow f_1 & & \downarrow f_0 & & \downarrow f_{-1} \\ \cdots \rightarrow C_3(L) & \xrightarrow{\partial_3} & C_2(L) & \xrightarrow{\partial_2} & C_1(L) & \xrightarrow{\partial_1} & C_0(L) & \xrightarrow{\partial_0} & \{0\} \end{array}$$

Induced map

We can define a linear map between quotient vector spaces:

$$(f_n)_* : Z_n(K) / B_n(K) \longrightarrow Z_n(L) / B_n(L).$$

By definition of the homology groups, we have defined a map

$$(f_n)_* : H_n(K) \longrightarrow H_n(L).$$

which is called the induced map in homology.

$$\begin{array}{ccccccc} \cdots & & H_3(K) & & H_2(K) & & H_1(K) & & H_0(K) \\ & & \downarrow (f_3)_* & & \downarrow (f_2)_* & & \downarrow (f_1)_* & & \downarrow (f_0)_* \\ \cdots & & H_3(L) & & H_2(L) & & H_1(L) & & H_0(L) \end{array}$$

Induced map

Example: Consider the simplicial complexes $K = \{[0], [1], [2], [0, 1], [0, 2], [1, 2]\}$ and $L = \{[0], [1], [2], [3], [0, 1], [0, 2], [1, 2], [1, 3], [2, 3]\}$.

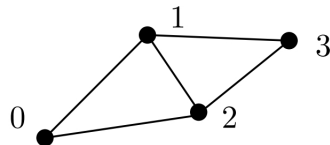
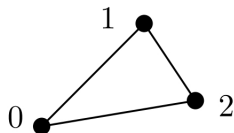
The homology group $H_1(L)$ is isomorphic to the vector space $(\mathbb{Z}/2\mathbb{Z})^2$ by identifying $[0, 1] + [0, 2] + [1, 2] \rightarrow (1, 0)$ and $[1, 2] + [2, 3] + [1, 3] \rightarrow (0, 1)$.

The inclusion $i: K \rightarrow L$ induces the following map between first homology groups:

$$(i_1)_* : H_1(K) \simeq \mathbb{Z}/2\mathbb{Z} \rightarrow H_1(L) \simeq (\mathbb{Z}/2\mathbb{Z})^2$$

$$1 \mapsto (1, 0)$$

It can be represented as the matrix $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$



Functor property

Let K, L, M be three simplicial complexes, and consider two simplicial maps $f: K \rightarrow L$ and $g: L \rightarrow M$.

For any $n \geq 0$, the induced map $((g \circ f)_n)_*: H_n(K) \rightarrow H_n(M)$ and $(g_n)_* \circ (f_n)_*: H_n(K) \rightarrow H_n(M)$ are equal.

$$\begin{array}{ccccc} & & g \circ f & & \\ & \nearrow & & \searrow & \\ K & \xrightarrow{f} & L & \xrightarrow{g} & M, \end{array}$$

$$\begin{array}{ccccc} & & (g \circ f)_* & & \\ & \nearrow & & \searrow & \\ H_n(K) & \xrightarrow{f_*} & H_n(L) & \xrightarrow{g_*} & H_n(M). \end{array}$$

Tracking cycles over time

Let $X \subset \mathbb{R}^n$. The collection of its thickenings is an non-decreasing sequence of subsets

$$\dots \subset X^{t_1} \subset X^{t_2} \subset X^{t_3} \dots$$

By considering the corresponding for example Čech complexes, we obtain an non-decreasing sequence of simplicial complexes

$$\dots \subset \text{Čech}^{t_1}(X) \subset \text{Čech}^{t_2}(X) \subset \text{Čech}^{t_3}(X) \subset \dots$$

Let us denote i_s^t the inclusion map corresponding to $\text{Čech}^s(X) \subset \text{Čech}^t(X)$. we can write

$$\text{-----} \rightarrow \check{\text{Cech}}^{t_1}(X) \xrightarrow{i_{t_1}^{t_2}} \check{\text{Cech}}^{t_2}(X) \xrightarrow{i_{t_2}^{t_3}} \check{\text{Cech}}^{t_3}(X) \text{-----}$$

Applying the i -th homology functor yields a diagram of vector spaces

$$\text{-----} \rightarrow H_i(\check{\text{Cech}}^{t_1}(X)) \xrightarrow{(i_{t_1}^{t_2})_*} H_i(\check{\text{Cech}}^{t_2}(X)) \xrightarrow{(i_{t_2}^{t_3})_*} H_i(\check{\text{Cech}}^{t_3}(X)) \text{-----}$$

where the maps $(i_s^t)_*$ are those induced in homology by the inclusions i_s^t .

Tracking cycles over time

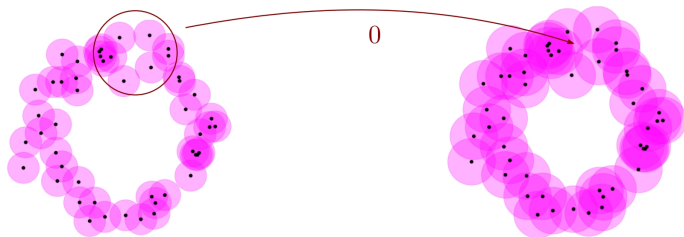
$$\text{-----} \rightarrow H_i(\check{\text{Cech}}^{t_1}(X)) \xrightarrow{(i_{t_1}^{t_2})_*} H_i(\check{\text{Cech}}^{t_2}(X)) \xrightarrow{(i_{t_2}^{t_3})_*} H_i(\check{\text{Cech}}^{t_3}(X)) \text{-----}$$

Let $i \geq 0$, $t_0 \geq 0$ and consider a cycle $c \in H_i(\text{Cech}^{t_0}(X))$.

Its death time is: $\sup\{t \geq t_0, (i_{t_0}^t)(c) \neq 0\}$,

Its birth time is: $\inf\{t \geq t_0, (i_t^{t_0})^{-1}(c) \neq \emptyset\}$

We measure the *persistence* of c as the difference between its death time and its birth time.



Tracking cycles over time

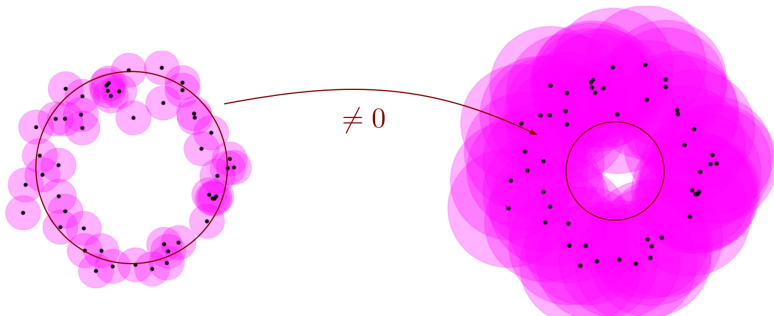
$$\text{-----} \rightarrow H_i(\check{\text{Cech}}^{t_1}(X)) \xrightarrow{(i_{t_1}^{t_2})_*} H_i(\check{\text{Cech}}^{t_2}(X)) \xrightarrow{(i_{t_2}^{t_3})_*} H_i(\check{\text{Cech}}^{t_3}(X)) \text{-----}$$

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We measure the *persistence* of c as the difference between its death time and its birth time.



persistence modules

Definition: A persistence module $\mathbb{V}$ over $\mathbb{R}^+$ with coefficients in $\mathbb{Z}/2\mathbb{Z}$ (generally in $\mathbb{F}$) is a pair $(\mathbb{V}, \nu)$ where $\mathbb{V} = (V^t)_{t \in \mathbb{R}^+}$ is a family of $\mathbb{Z}/2\mathbb{Z}$ -vector spaces, and $\nu = (\nu_s^t: V^s \rightarrow V^t)_{s \leq t \in \mathbb{R}^+}$ a family of linear maps such that:

- ▶ for every $t \in \mathbb{R}^+$, $\nu_t^t: V^t \rightarrow V^t$ is the identity map,
- ▶ for every $r, s, t \in \mathbb{R}^+$ such that $r \leq s \leq t$, we have $\nu_s^t \circ \nu_r^s = \nu_r^t$.

In practice, one builds persistence modules from filtrations.

Definition: An isomorphism between two persistence modules $\mathbb{V}$ and $\mathbb{W}$ is a family of isomorphisms of vector spaces $\phi = (\phi_t: V^t \rightarrow W^t)_{t \in \mathbb{R}^+}$ such that the following diagram commutes for every $s \leq t \in \mathbb{R}^+$:

$$\begin{array}{ccc} V^s & \xrightarrow{\nu_s^t} & V^t \\ \downarrow \phi_s & & \downarrow \phi_t \\ W^s & \xrightarrow{w_s^t} & W^t \end{array}$$

Decomposability

Definition: Let $(\mathbb{V}, v)$ and $(\mathbb{W}, w)$ be two persistence modules. Their sum is the persistence module $\mathbb{V} \oplus \mathbb{W}$ defined with the vector spaces $(V \oplus W)^t = V^t \oplus W^t$ and the linear maps

$$(v \oplus w)_s^t : (x, y) \in (V \oplus W)^s \longmapsto (v_s^t(x), w_s^t(y)) \in (V \oplus W)^t.$$

A persistence module $\mathbb{U}$ is indecomposable if for every pair of persistence modules $\mathbb{V}$ and $\mathbb{W}$ such that $\mathbb{U}$ is isomorphic to the sum $\mathbb{V} \oplus \mathbb{W}$, then one of the summands has to be a trivial persistence module, that is, equal to zero for every $t \in \mathbb{R}^+$.

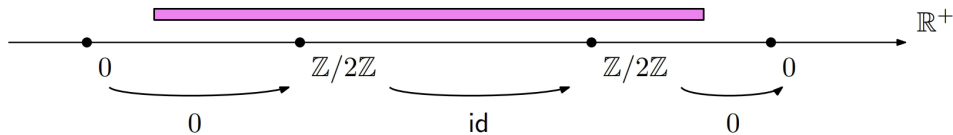
Otherwise, $\mathbb{U}$ is said decomposable.

Definition: Let $I \subset \mathbb{R}^+$ be an interval: $[a, b]$, $(a, b]$, $[a, b)$ or (a, b) , with $a, b \in \mathbb{R}^+$ such that $a \leq b$, and potentially $a = -\infty$ or $b = +\infty$.

Interval Module

Definition: The interval module associated to I is the persistence module $\mathbb{B}[I]$ with vector spaces $\mathbb{B}^t[I]$ and linear maps $v_s^t: \mathbb{B}^s[I] \rightarrow \mathbb{B}^t[I]$ defined as

$$\mathbb{B}^t[I] = \begin{cases} \mathbb{Z}/2\mathbb{Z}(\mathbb{F}) & \text{if } t \in I, \\ 0 & \text{otherwise,} \end{cases} \quad \text{and} \quad v_s^t = \begin{cases} \text{id} & \text{if } s, t \in I, \\ 0 & \text{otherwise.} \end{cases}$$



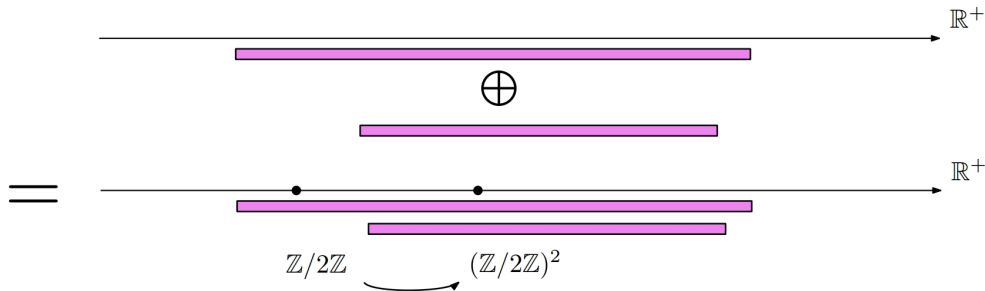
Lemma: Interval modules are indecomposable.

Interval Modules

Definition: The interval module associated to I is the persistence module $\mathbb{B}[I]$ with vector spaces $\mathbb{B}^t[I]$ and linear maps $\nu_s^t: \mathbb{B}^s[I] \rightarrow \mathbb{B}^t[I]$ defined as

$$\mathbb{B}^t[I] = \begin{cases} \mathbb{Z}/2\mathbb{Z} & \text{if } t \in I, \\ 0 & \text{otherwise,} \end{cases} \quad \text{and} \quad \nu_s^t = \begin{cases} \text{id} & \text{if } s, t \in I, \\ 0 & \text{otherwise.} \end{cases}$$

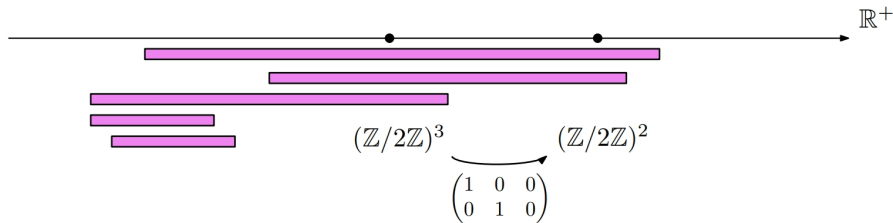
We can sum interval modules:



Interval Modules

Definition: The interval module associated to I is the persistence module $\mathbb{B}[I]$ with vector spaces $\mathbb{B}^t[I]$ and linear maps $\nu_s^t: \mathbb{B}^s[I] \rightarrow \mathbb{B}^t[I]$ defined as

$$\mathbb{B}^t[I] = \begin{cases} \mathbb{Z}/2\mathbb{Z} & \text{if } t \in I, \\ 0 & \text{otherwise,} \end{cases} \quad \text{and} \quad \nu_s^t = \begin{cases} \text{id} & \text{if } s, t \in I, \\ 0 & \text{otherwise.} \end{cases}$$



Persistence barcodes and persistence diagrams

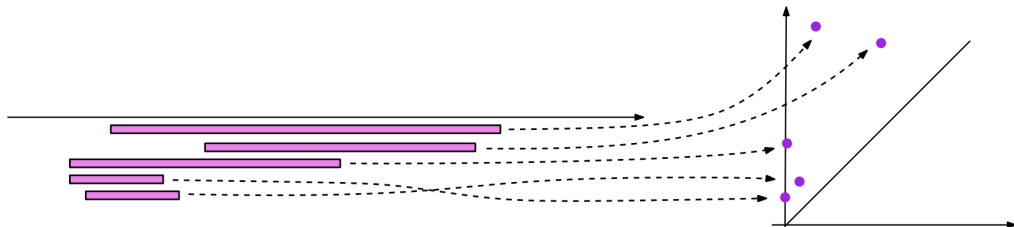
Definition: A persistence module $\mathbb{V}$ decomposes into interval module if there exists a set $\{\mathbb{B}_i, i \in \mathcal{I}\}$ of interval modules such that $\mathbb{V}$ is isomorphic to the sum $\bigoplus_{i \in \mathcal{I}} \mathbb{B}_i$. In other words, there exists a multiset $\mathcal{I}$ of intervals of T such that

$$\mathbb{V} \simeq \bigoplus_{I \in \mathcal{I}} \mathbb{B}[I]$$

Multiset means that $\mathcal{I}$ may contain several copies of the same interval I .

In this case, $\mathcal{I}$ is called the persistence barcode of $\mathbb{V}$. It is written $\text{Barcode}(\mathbb{V})$.

For every $[a, b], (a, b], [a, b)$ or (a, b) in $\text{Barcode}(\mathbb{V})$, consider the point (a, b) of $\mathbb{R}^2$. The collection of all such points is the *persistence diagram* of $\mathbb{V}$.



Persistent Homology algorithm

After considering the boundary matrix, and computing a Gauss reduction. Now, for all j such that $\delta(j)$ is defined, consider the pair of simplices

$$(\sigma^{\delta(j)}, \sigma^j).$$

Also, for all i such that $\forall j, \delta(j) \neq i$, we set: $(\sigma^i, +\infty)$.

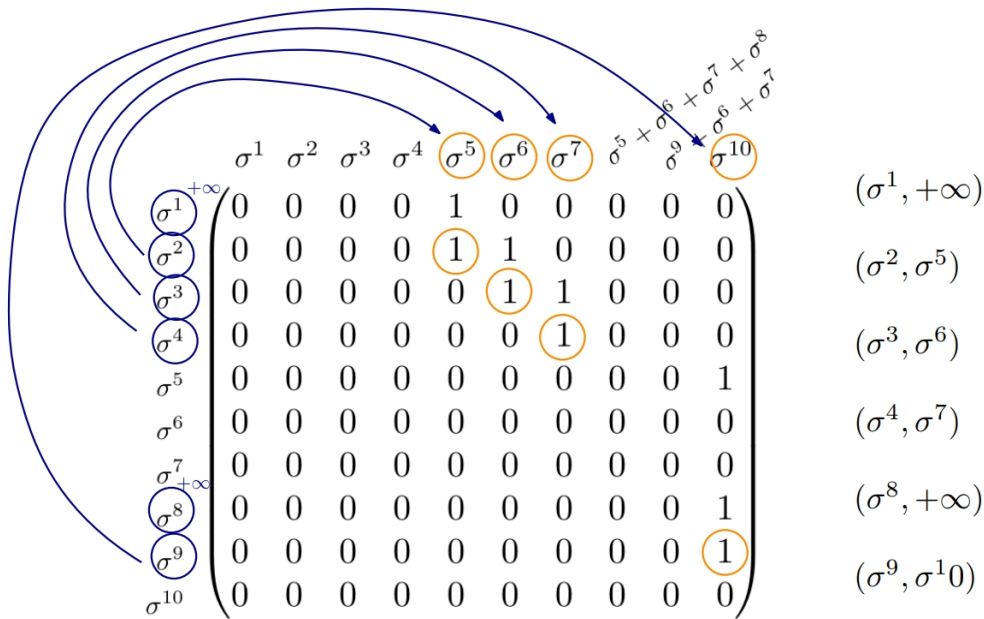
The pairs of simplices (σ, τ) are called *persistence pairs*.

We do that because from these persistence pairs we can read the barcode of our persistence modules.

Proposition: The barcodes of the filtration consists in the intervals

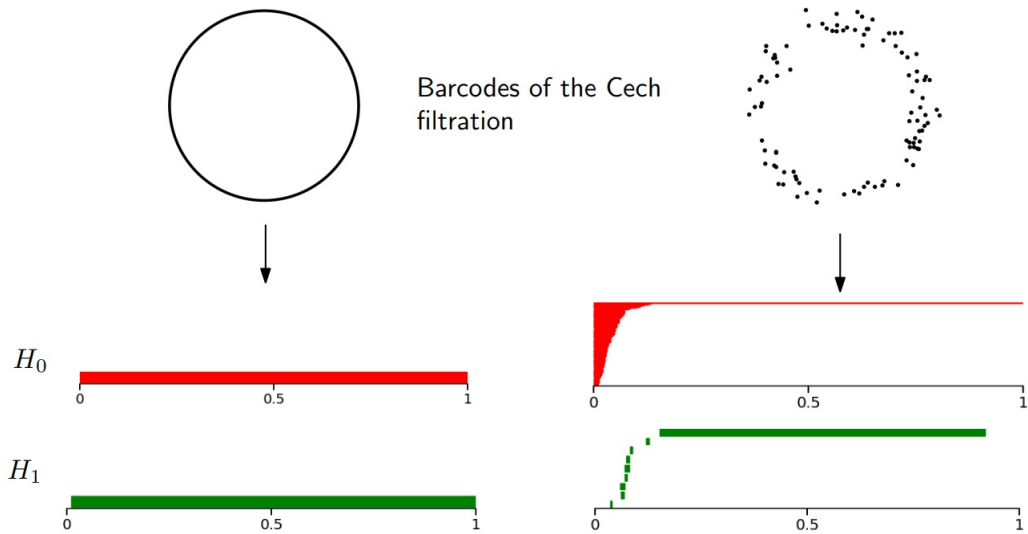
$$\mathcal{J} = \{(t(\sigma), t(\tau)) \text{ for all persistence pair } (\sigma, \tau) \text{ such that } t(\sigma) \neq t(\tau)\}$$

Denote $t(\sigma)$ the time of apparition of the simplex σ in the filtration.



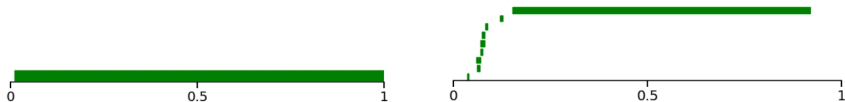
Distances between persistence modules

Let $X \subset \mathbb{R}^n$, seen as a sample of M (In this case nice theoretical circle).



Matching

Consider two barcodes P and Q , that is, multisets of intervals $\{(a_i, b_i), i \in \mathcal{I}\}$ of such that $a_i \leq b_i$ for all $i \in \mathcal{I}$.



A **partial matching** between the barcodes is a subset $M \subset P \times Q$ such that

- ▶ for every $p \in P$, there exists at most one $q \in Q$ such that $(p, q) \in M$,
- ▶ for every $q \in Q$, there exists at most one $p \in P$ such that $(p, q) \in M$.

The points $p \in P$ (resp. $q \in Q$) such that there exists $q \in Q$ (resp. $p \in P$) with $(p, q) \in M$ are said **matched** by M .

If a point $p \in P$ (resp. $q \in Q$) is not matched by M , we consider that it is matched with the singleton $\bar{p} = [\frac{p_1+p_2}{2}, \frac{p_1+p_2}{2}]$ (resp. $\bar{q} = [\frac{q_1+q_2}{2}, \frac{q_1+q_2}{2}]$).

The **cost** of a matched pair (p, q) (resp. $(p, \bar{p})$, resp. $(\bar{q}, q)$) is the sup norm

$$\|p - q\|_{\infty} = \sup\{|p_1 - q_1|, |p_2 - q_2|\} \quad (\text{resp. } \|p - \bar{p}\|_{\infty}, \text{ resp. } \|\bar{q} - q\|_{\infty}).$$

The **cost** of the partial matching M , denoted $\text{cost}(M)$, is the supremum of all costs.

Bottleneck distance

Definition: The bottleneck distance between the bars P and Q is defined as the infimum of costs over all the partial matchings:

$$d_b(P, Q) = \inf\{\text{cost}(M), M \text{ is a partial matching between } P \text{ and } Q\}$$

If $\mathbb{U}$ and $\mathbb{V}$ are two decomposable persistence modules, we define their bottleneck distance as

$$d_b(\mathbb{U}, \mathbb{V}) = d_b(\text{Diagram}(\mathbb{U}), \text{Diagram}(\mathbb{V})).$$

Example: Let $a, a', b, b' \in \mathbb{R}^+$ such that $a \leq b$ and $a' \leq b'$. Consider the interval-modules $\mathbb{B}[a, b]$ and $\mathbb{B}[a', b']$.
we deduce

$$d_b(\mathbb{B}[a, b], \mathbb{B}[a', b']) = \min \left\{ \max \left\{ \frac{b-a}{2}, \frac{b'-a'}{2} \right\}, \max \{|a-a'|, |b-b'|\} \right\}.$$

Interleaving distance

Consider two persistence modules $\mathbb{V}$ and $\mathbb{W}$:

$$\begin{array}{ccccccc} \cdots \rightarrow & V^{t_1} & \xrightarrow{v_{t_1}^{t_2}} & V^{t_2} & \xrightarrow{v_{t_2}^{t_3}} & V^{t_3} & \xrightarrow{v_{t_3}^{t_4}} & V^{t_4} & \cdots \\ \cdots \rightarrow & W^{t_1} & \xrightarrow{w_{t_1}^{t_2}} & W^{t_2} & \xrightarrow{w_{t_2}^{t_3}} & W^{t_3} & \xrightarrow{w_{t_3}^{t_4}} & W^{t_4} & \cdots \end{array}$$

Given $\epsilon \geq 0$, an ϵ -shift (ϵ -morphism) between $\mathbb{V}$ and $\mathbb{W}$ is a family of linear maps $\phi = (\phi_t: V^t \rightarrow W^{t+\epsilon})_{t \in \mathbb{R}^+}$ such that the following diagram commutes for every $s \leq t \in \mathbb{R}^+$:

$$\begin{array}{ccc} V^s & \xrightarrow{v_s^t} & V^t \\ \downarrow \phi_s & & \downarrow \phi_t \\ W^{s+\epsilon} & \xrightarrow{w_{s+\epsilon}^{t+\epsilon}} & W^{t+\epsilon} \end{array}$$

Interleaving distance

An ϵ -interleaving between $\mathbb{V}$ and $\mathbb{W}$ is a pair of ϵ -morphisms $(\phi_t: V^t \rightarrow W^{t+\epsilon})_{t \in \mathbb{R}^+}$ and $(\psi_t: W^t \rightarrow V^{t+\epsilon})_{t \in \mathbb{R}^+}$ such that the following diagrams commute for every $t \in \mathbb{R}^+$:

$$\begin{array}{ccc} V^t & \xrightarrow{v_t^{t+2\epsilon}} & V^{t+2\epsilon} \\ & \searrow \phi_t \quad \nearrow \psi_{t+\epsilon} & \\ & W^{t+\epsilon} & \end{array}$$

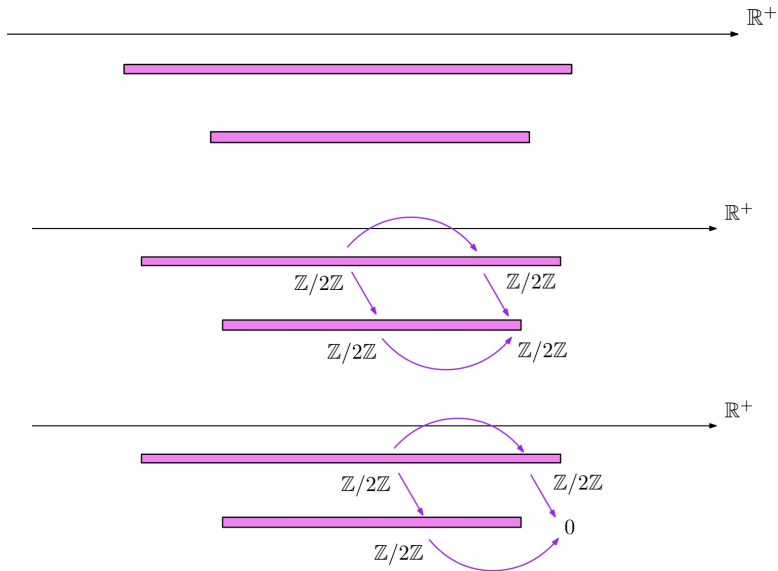
$$\begin{array}{ccc} & V^{t+\epsilon} & \\ \psi_t \nearrow & & \searrow \phi_{t+\epsilon} \\ W^t & \xrightarrow{w_t^{t+2\epsilon}} & W^{t+2\epsilon} \end{array}$$

The interleaving distance is:

$$d_i(\mathbb{V}, \mathbb{W}) = \inf\{\epsilon \geq 0, \mathbb{V} \text{ and } \mathbb{W} \text{ are } \epsilon\text{-interleaved}\}.$$

Example: Let $a, a', b, b' \in \mathbb{R}^+$ such that $a \leq b$ and $a' \leq b'$. Consider the interval-modules $\mathbb{B}[a, b]$ and $\mathbb{B}[a', b']$.

Let's find an ϵ -interleaving.

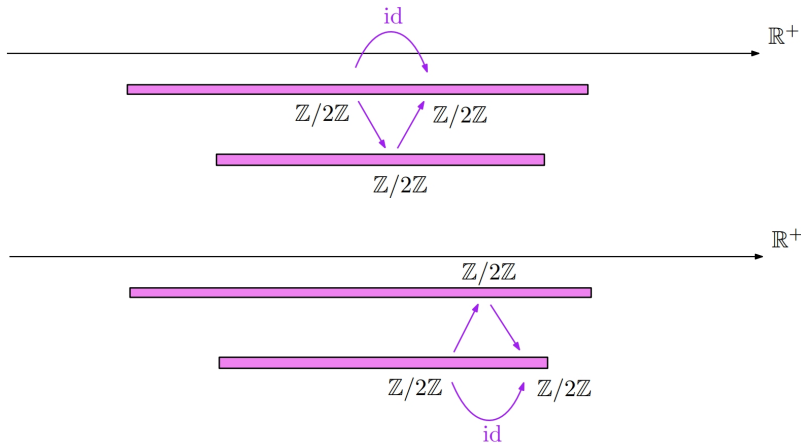


only two possibilities for ϕ :

always the zero map OR always nonzero when V^t and $W^{t+\epsilon}$ are nonzero.

$\psi_{t+\epsilon} \circ \phi_t$ must be nonzero when $[t, t+\epsilon] \subset [a, b]$.

$\phi_{t+\epsilon} \circ \psi_t$ must be nonzero when $[t, t+\epsilon] \subset [a', b']$.



We deduce that either

- ▶ $|a - b| \leq 2\epsilon$ and $|a' - b'| \leq 2\epsilon$, or
- ▶ $|a - a'| \leq \epsilon$ and $|b - b'| \leq \epsilon$

Conclusion:

$$d_i(\mathbb{B}[a, b], \mathbb{B}[a', b']) = \min \left\{ \max \left\{ \frac{b-a}{2}, \frac{b'-a'}{2} \right\}, \max \{|a-a'|, |b-b'|\} \right\}.$$

Back to Thickenings

Let X and Y be two subsets of $\mathbb{R}^n$. Define $\epsilon = d_H(X, Y)$ (Hausdorff distance).

We have seen that $X \subset Y^\epsilon$ and $Y \subset X^\epsilon$.

We even have that $X^t \subset Y^{t+\epsilon}$ and $Y^t \subset X^{t+\epsilon}$ for all $t \geq 0$.

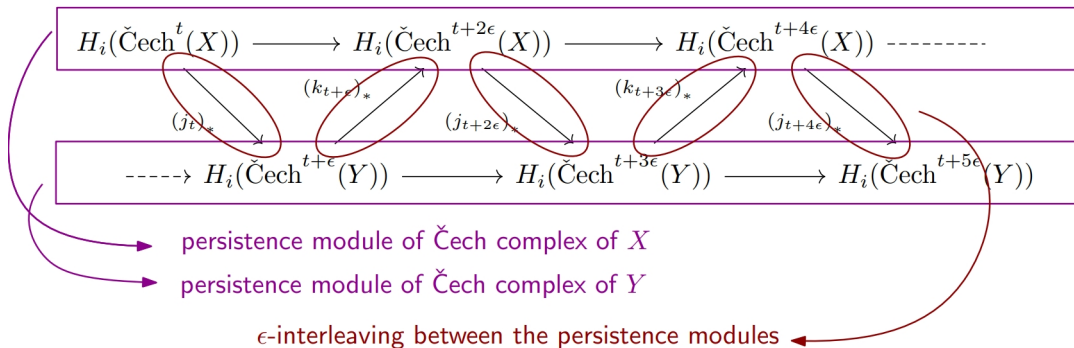
By denoting j and k these inclusions, we have a commutative diagram

$$\begin{array}{ccccccc}
 \cdots \rightarrow & X^t & \xrightarrow{\quad} & X^{t+2\epsilon} & \xrightarrow{\quad} & X^{t+4\epsilon} & \leftarrow \cdots \\
 & \searrow j_t & & \nearrow k_{t+\epsilon} & & \nearrow k_{t+3\epsilon} & \\
 & & & & \searrow j_{t+2\epsilon} & & \searrow j_{t+4\epsilon} \\
 \cdots \rightarrow & Y^{t+\epsilon} & \xrightarrow{\quad} & Y^{t+3\epsilon} & \xrightarrow{\quad} & Y^{t+5\epsilon} & \leftarrow \cdots
 \end{array}$$

This also gives inclusions between Čech complexes:

$$\begin{array}{ccccccc}
 \text{Čech}^t(X) & \xrightarrow{\quad} & \text{Čech}^{t+2\epsilon}(X) & \xrightarrow{\quad} & \text{Čech}^{t+4\epsilon}(X) & \leftarrow \cdots \\
 & \searrow j_t & & \nearrow k_{t+\epsilon} & & \nearrow k_{t+3\epsilon} & \\
 & & & & \searrow j_{t+2\epsilon} & & \searrow j_{t+4\epsilon} \\
 \cdots \rightarrow & \text{Čech}^{t+\epsilon}(Y) & \xrightarrow{\quad} & \text{Čech}^{t+3\epsilon}(Y) & \xrightarrow{\quad} & \text{Čech}^{t+5\epsilon}(Y) & \leftarrow \cdots
 \end{array}$$

Now, we apply the i -th homology functor.



Hence the persistence modules $(H_i(\text{Cech}^t(X)))_{t \geq 0}$ and $(H_i(\text{Cech}^t(Y)))_{t \geq 0}$ are ϵ -interleaved.

Stability theorem

Hence $d_i(\mathbb{U}, \mathbb{V}) \leq \epsilon$.

Theorem

If the persistence modules $\mathbb{U}$ and $\mathbb{V}$ are interval-decomposable, then

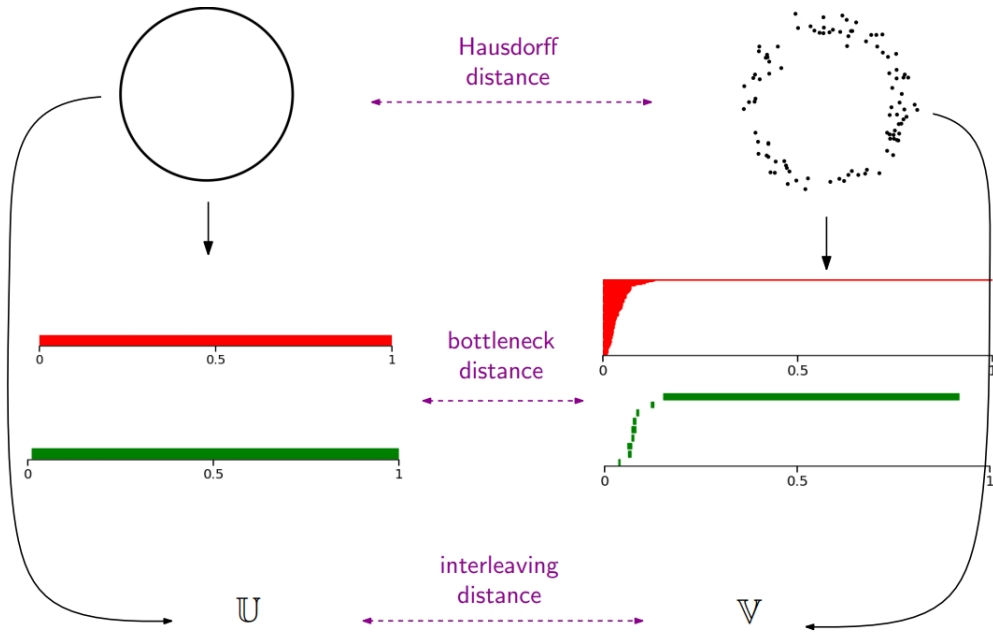
$$d_i(\mathbb{U}, \mathbb{V}) = d_b(\mathbb{U}, \mathbb{V})$$

We use isometry theorem: $d_b(\mathbb{U}, \mathbb{V}) = d_i(\mathbb{U}, \mathbb{V}) \leq \epsilon$

Theorem

Let X and Y be two subsets of $\mathbb{R}^n$. consider their Cech (resp. Rips) filtration, and the corresponding i -th homology persistence modules, $\mathbb{U}$ and $\mathbb{V}$. Suppose that they are interval-decomposables. Then

$$d_b(\mathbb{U}, \mathbb{V}) \leq d_H(\mathbb{U}, \mathbb{V}).$$



ZigZag persistence

Consider a zigzag persistence module V of the form

$$V: V_1 \xleftrightarrow{f_1} V_2 \xleftrightarrow{f_2} \dots \xleftrightarrow{f_{n_2}} V_{n-1} \xleftrightarrow{f_{n-1}} V_n.$$

If $f_i: V_i \longrightarrow V_{i+1}$, then we say that f_i is forward, and if $f_i: V_i \longleftarrow V_{i+1}$, then we say that f is backward.

Definition: A representative sequence in V is an n -tuple

$$v = (v_1, v_2, \dots, v_n) \in V_1 \times V_2 \times \dots \times V_n$$

such that:

1. **The index** $b(v)$:

- ▶ $b(v)$ is the smallest index where $v_i \neq 0$.
- ▶ $b(v) = 1$ if either:
 - ▶ The map $f_{b(v)-1}$ is forward, or
 - ▶ $f_{b(v)-1}$ is backward, and applying the backward map to $v_{b(v)}$ gives 0, i.e., $f_{b(v)-1}(v_{b(v)}) = 0$.
- ▶ For all indices i less than $b(v)$, $v_i = 0$.

2. The index $d(v)$:

- ▶ $d(v)$ is the largest index where $v_i \neq 0$.
- ▶ $d(v) = n$ if either:
 - ▶ The map $f_{d(v)}$ is backward, or
 - ▶ $f_{d(v)}$ is forward and applying the forward map to $v_{d(v)}$ gives 0, i.e., $f_{d(v)}(v_{d(v)}) = 0$.
- ▶ For all indices i greater than $d(v)$, $v_i = 0$.

3. **Non-zero condition:** For all i between $b(v)$ and $d(v)$, $v_i \neq 0$.

4. **Forward maps:** If $b(v) \leq i < d(v)$ and the map f_i is forward, then $f_i(v_i) = v_{i+1}$.

5. **Backward maps:** If $b(v) < i \leq d(v)$ and the map f_i is backward, then $f_i(v_i) = v_{i-1}$.

we assume that K is a zigzag of simplicial complexes

$$K: K_1 \xleftrightarrow{h_1} K_2 \xleftrightarrow{h_2} \dots \xleftrightarrow{h_{n-2}} K_{n-1} \xleftrightarrow{h_{n-1}} K_n.$$

where each h_i is an inclusion, and any two consecutive simplicial complexes differ by exactly one simplex. This assumption implies that the induced map

$$(h_i)_* : H_p(K_i) \longleftrightarrow H_p(K_{i+1})$$

is either injective, surjective or an isomorphism, and that the dimension between consecutive vector spaces differ by at most 1.

Let $H_p(K)$ denote the zigzag module

$$H_p(k_1) \leftrightarrow H_p(k_2) \leftrightarrow \dots \leftrightarrow H_p(k_{n-1}) \leftrightarrow H_p(k_n)$$

Definition: The representative cycle c is defined as an n -tuple $c = (c_1, \dots, c_n)$ belonging to the product of chain complexes $C_p(K_1) \times \dots \times C_p(K_n)$, where $C_p(K_i)$ is the chain group of p -simplices for the complex K_i .

Such that The cycle $c = (c_1, \dots, c_n)$ satisfies:

$$0 \neq [c_i] \in H_p(K_i) \text{ if only if } c_i \neq 0,$$

and

$$[c] = ([c_1], \dots, [c_n])$$

is a representative sequence in $H_p(K)$.

The goal is to construct a family of representative cycles $\{c^j\}_{j \in J}$ that forms a basis for $H_p(K)$, i.e., a basis for the zigzag homology.

Inductive construction:

The cycles are constructed inductively, meaning that the base case $k = 1$ is trivial. The set $\Sigma_i = \{c^j\}_{j \in J}$ is a family for $H_p(K)$ restricted to $\{1, 2, \dots, i\}$.

Birth of Cycles

The elements of the index set J are ordered by their birth function $b(c^j)$.

- ▶ Forward birth occurs when $b([c^j]) = 1$ or when $f_{b(c^j)-1}$.
- ▶ Backward birth occurs when $b([c^j]) \geq 1$ and the map before it is backward.

Extension Process:

The family Σ^i is extended to Σ^{i+1} based on a case-by-case analysis:

- ▶ If $(h_i)_*$ is forward and an isomorphism, the extension is straightforward, just adding a new cycle.
- ▶ If $(h_i)_*$ is backward and an isomorphism, the extension also just adds a new cycle.
- ▶ If $(h_i)_*$ is forward and injective, a cycle c' is created with the addition of a new simplex.
- ▶ If $(h_i)_*$ is backward and surjective, a cycle is extended by adding a new simplex related to a boundary simplex $\partial_{p+1}(\sigma)$.

$(h_i)_*$ is Forward and Surjective:

1. Set Definition:

- ▶ There is a set $A \subseteq J$ such that the sum $\sum_{j \in A} [c_i^j] = 0$, where $[c_i^j] \neq 0$ for all $j \in A$.
- ▶ In other words, there is a collection of cycles whose homology classes sum to zero at the i -th complex.

2. Birth Classification:

- ▶ If $b([c^j])$ (the birth function of the cycle) is a *backward birth* for all $j \in A$, let λ be the smallest index in A .
- ▶ Otherwise, let λ be the largest index in A such that $b([c^\lambda])$ is a *forward birth*.

3. Extension:

- ▶ The family Σ^{i+1} is extended by adding a new cycle c' to all the elements in Σ^i , with c' defined based on the birth times:
 - ▶ $c'_x = 0$ for all $x < b(v^\lambda)$ and for $x = i + 1$.
 - ▶ $c'_x = \sum_{j \in A} c_x^j$ otherwise.

$(h_i)_*$ is backward and Injective:

1. Set Definition:

- ▶ Define a set $A \subseteq J$ such that c_i^j contains the p -simplex σ , where $K_i = K_{i+1} \cup \{\sigma\}$

2. Birth Classification:

- ▶ If $b([c^j])$ (the birth function of the cycle) is a *forward birth* for all $j \in A$, let λ be the smallest index in A .
- ▶ Otherwise, let λ be the largest index in A such that $b([c^\lambda])$ is a *backward birth*.

3. Extension:

- ▶ The family Σ^{i+1} is extended by adding a new cycle c' in two parts:
 - ▶ for $j \notin A$, extend the cycle c^j .
 - ▶ Add a new cycle $(c^1, \dots, c^\lambda, 0)$ corresponding to the backward birth and a simplex σ .

The cycle c' for $j \in A$ is defined as follows:

- ▶ $(c')_x = c_x^j + c_x^\lambda$ for $b([c^\lambda]) \leq x \leq i$.
- ▶ $(c')_{i+1} = (c^j)_i$, and otherwise it remains c_x^j .

Thank You!